

Guest

Browses and searches public reports and the map

Customer / User

Files lost and found reports, reviews possible matches

Staff / Pet Coordinator

Verifies pairings, coordinates the handover

Administrator

Accounts, roles, categories, moderation



HTTPS · one origin: the site and the API are served together

PRESENTATION

in the browser

React 19.2

Vite 8.2

Tailwind CSS 4.3

React Router 7.18

Leaflet 1.9 + OpenStreetMap

Displays state

Pages ask the API for everything through one service layer. Hiding a button is never the security: every action the page offers is checked again by the server.



Fetch / AJAX · JSON · PHP session cookie (HttpOnly, SameSite=Lax, Secure) · CSRF token header on every change

APPLICATION

on the server

PHP 8.3 REST API · Apache · JSON in, JSON out · our own endpoints

Authentication

bcrypt hashes, sessions, lock after 3 failed sign-ins, email verification

Authorization

role re-read from the database on every request; 401 / 403

Validation

every field checked on the server; 422 names the field

Privacy filtering

unpublished contact never leaves the server; guests get a summary

Rule-based matching

7 weighted signals, suggestion at 65+ of 100; no AI

Notifications

in-app, stored per user; email only for account links

Moderation

anyone signed in flags; an administrator decides

Audit log

sign-ins, role changes, decisions: who, what, when

Enforces rules



PDO · native prepared statements (emulation off) · transactions for multi-step changes

DATA

Railway MySQL service

Owens persistent state

MySQL 9.4

InnoDB · utf8mb4

17

physical tables

15

domain tables

24

foreign keys

15 on the ERD. The other two, schema_migrations and auth_rate_limits, are operational.

PRODUCTION · RAILWAY

Application service

One Docker container: Apache + PHP 8.3 serving the built React site and /api from one origin. Health check /api/health.

MySQL 9.4 service

Separate service; the app reads its address and credentials from environment variables.

Persistent volume

/var/lib/pawsandfound: uploaded photos and PHP sessions, kept across redeployments.

EXTERNAL SERVICES

Brevo HTTPS Mail API

Email verification, password reset and email-change links.

Cloudflare Turnstile

Bot check on registration.

OpenStreetMap tiles

Map imagery, loaded by the browser through Leaflet.

ONE REQUEST, END TO END: FILING A REPORT

1. React form → POST /api/reports, JSON, with the session cookie and CSRF token
2. PHP checks the CSRF token, then that someone is signed in and active
3. Every field validated again; a bad one is a 422 naming it
4. Location, report and first history line saved in **one transaction**
5. After the commit, matching compares it with open reports and notifies both reporters

LOCAL DEVELOPMENT ONLY, NOT PRODUCTION

XAMPP (Apache + PHP, MariaDB 10.4 on port 3307) and the Vite dev server. Same code, same database/schema.sql.

React displays state. PHP enforces rules. MySQL owns persistent state.